



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

01-07-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE-II)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,C,D	A,B	A,C	A,B	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	B	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.41	22.50	6.00	12.00	3.24	8.00	12.00	32.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	A,B,C,D	C,D	A,B,D	A,B,C,D	A,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	C	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	19.00	5.00	9.00	11.00	3.00	8.40	500.00	63.18

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,D	A,B,C,D	B,C,D	A,B,D	B,C,D	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	A	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.71 or 2.72	8.00	2.66 or 2.67	1.00	0.00	0.00	13.00	5.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

2. Ans (A,C,D)

At any instant $T_{\text{spring}} = T_{\text{string}}$

($\because$ Massless, frictionless)

$\therefore$ At any instant tension in both springs will be same.

3. Ans (A,B)

$$1 \text{ msd} - 1 \text{ vsd} = 1 - \frac{19}{20} \text{ mm} = \frac{1}{20} \text{ mm}$$

$$d = 8.2 \text{ cm} + \frac{6}{20} \text{ mm}$$

$$d_{\text{outer}} = 3.4 \text{ cm} + \frac{16}{20} \text{ mm}$$

$$d_{\text{in}} = 3.4 \text{ cm} + \frac{4}{20} \text{ mm}$$

4. Ans (A,C)

For smaller sphere, before

(covering $P_{\text{point source}} = P_{\text{radiated}}$)

$$\therefore P = \sigma (4\pi R^2) T_0^4$$

After covering,

for outer shell : $P_{\text{point source}} = P_{\text{radiated}}$

$$\therefore \sigma (4\pi R^2) T_0^4 = \sigma (4\pi (2R)^2) T^4$$

$$\therefore T_0^4 = 4T^4 \Rightarrow T = \frac{T_0}{\sqrt{2}}$$

After covering, temperature of inner sphere will be greater than T_0

5. Ans (A,B)

$$f_{\text{lim}} = \mu_s N = 0.25 \times 40 = 10 \text{ N}$$

$\therefore$ Block starts moving at $t > 2 \text{ sec}$

$$\text{For } t > 2, a = \frac{F - 10}{m} = \frac{F - 10}{4}$$

Maximum speed is acquired at 6 sec as after $t = 6 \text{ sec}$, $F < 10 \text{ N}$

6. Ans (A,B,C)

$$[\phi] = \frac{[h]}{[\sigma\theta^4]} = \frac{ML^2T^{-1}}{MT^{-3}K^{-4}K^4} = L^2T^2$$

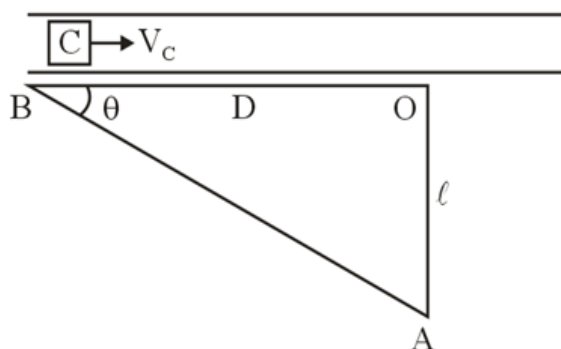
$$\frac{(\text{Weber}) (\Omega)^2 (\text{Farad})^2}{\text{Tesla}}$$

$$= \frac{Tm^2\Omega^2F^2}{T} = m^2s^2$$

PART-1 : PHYSICS

SECTION-I (ii)

8. Ans (B)



Let the car be at point 'B' when it sends a signal that the detector detects at 900 Hz

$$900 = 600 \left[\frac{340}{340 - V_c \cos \theta} \right] \Rightarrow \cos \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ$$

By the time signal reaches 'A' the car reaches at D.

Let this time be t_0 .

$$\therefore t_0 = \frac{AB}{V_s} = \frac{l \cos \theta}{V_s}$$

Now man fires bullet and it reaches 'O' in time t_1 .

In the same time, car moves from D to O.

$$\therefore t_1 = \frac{l}{V_B} \quad (V_B = \text{Speed of Bullet})$$

$$\text{Also, } BD + DO = l \cot \theta \Rightarrow V_c t_0 + V_c t_1 = l \cot \theta$$

$$\Rightarrow V_c \left[\frac{l \cos \theta}{V_s} \right] + V_c \left[\frac{l}{V_B} \right] = l \cot \theta$$

$$\Rightarrow V_B = \frac{3\sqrt{3}}{5} V_c = \frac{3\sqrt{3}}{5} \times \frac{2}{3\sqrt{3}} V_{\text{sound}}$$

$$\Rightarrow V_B = \frac{2}{5} V_{\text{sound}} = \frac{2}{5} \times 340 = 136 \text{ m/sec}$$

Option C

Time taken by sound to reach shooter

$$\Rightarrow t_2 = \frac{l}{V_s}$$

$$\text{Distance covered by car} = V_c \times \frac{l}{V_s} = \frac{2l}{3\sqrt{3}}$$

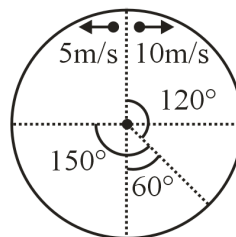
Option B

9. Ans (B)

Velocity of A after throwing = 5 m/s

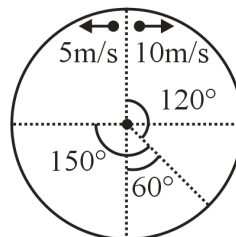
Total impulse = $40 \times 5 + 40 \times 5$

$$= 400 \text{ kg-m/sec}$$



10. Ans (A)

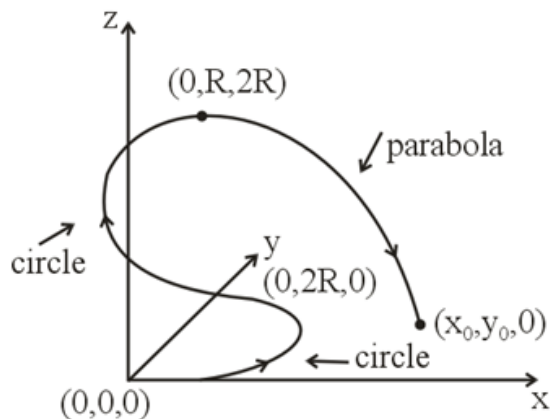
Angular displacement of A = $90^\circ + 60^\circ = 150^\circ$



PART-1 : PHYSICS

SECTION-II

1. Ans (1.41)



$$R = \frac{mv_0}{qB_0}$$

For parabolic path,

$$2R = \frac{1}{2} \left(\frac{qE_0}{m} \right) t^2$$

$$t = 2\sqrt{\frac{mR}{qE_0}} \therefore x_0 = V_0 t$$

$$= 2V_0 \sqrt{\frac{mR}{qE_0}} = 2V_0 \sqrt{\frac{m}{qE_0} \frac{mV_0}{qB_0}}$$

$$y_0 = 2R = \frac{2mv_0}{qB_0} \quad x_0 = \frac{2mv_0}{qB_0} \sqrt{\frac{B_0 V_0}{E_0}}$$

2. **Ans (22.50)**

$$T_{\text{string}} = 5t \text{ Newton}$$

1kg block loses contact at $t = 2\text{sec}$ & 2kg block loses contact at $t = 4\text{ sec}$.

After losing contact,

$$a_{1\text{kg}} = \left(\frac{5t - 10}{1} \right) \text{ m/s}^2$$

$$\therefore V_{1\text{kg}} \text{ at } t = 6 \text{ sec} = \int_2^6 a dt = \int_2^6 (5t - 10) dt = \left[\frac{5t^2}{2} - 10t \right]_2^6$$

$$= 40 \text{ m/sec}$$

$$V_{2\text{kg}} \text{ at } t = 6 \text{ sec} = \int_4^6 \left(\frac{5t - 20}{2} \right) dt = \left[\frac{5t^2}{4} - 10t \right]_4^6 = 5 \text{ m/sec}$$

$$\therefore V_{\text{pulley}} = \frac{40 + 5}{2} = 22.50 \text{ m/sec}$$

3. **Ans (6.00)**

For plano convex silvered lens:

$$\frac{1}{f_{\text{eq}}} = \frac{1}{f_m} + \frac{1}{f_l} = \frac{1}{-21} - \frac{2}{84} = \frac{-3}{42}$$

$$\therefore \frac{1}{f_l} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{\infty} - \frac{1}{-42} \right) = \frac{1}{84}$$

$$\Rightarrow f_{\text{eq}} = -14 \text{ cm}$$

i.e. it behaves like concave mirror of

$$\text{R.O.C.} = 28 \text{ cm.}$$

Shift due to transparent slab

$$= t \left[1 - \frac{1}{\mu} \right] = 6 \left[1 - \frac{2}{3} \right] = 2 \text{ cm}$$

For image on object : $(d + 24) - (\text{Shift}) = 28 \text{ cms}$

$$\Rightarrow (d + 24) - 2 = 28 \text{ cms}$$

$$\Rightarrow d = 6 \text{ cms.}$$

4. **Ans (12.00)**

$$\frac{0.1}{50} = \frac{h_1}{50} \Rightarrow h_1 = 0.1 \text{ cm}$$

$$\frac{0.1}{20} = \frac{L + h_1}{80} \Rightarrow L + h_1 = 0.4 \text{ cm} \Rightarrow L = 0.3 \text{ cm}$$

$$\beta = \frac{\lambda D}{d} = \frac{5000 \times 10^{-10}}{2 \times 10^{-3}} = 2.5 \times 10^{-4} \text{ m}$$

$$\therefore \text{No. of fringes} = \frac{L}{\beta} = \frac{3 \times 10^{-3}}{2.5 \times 10^{-4}} = 12$$

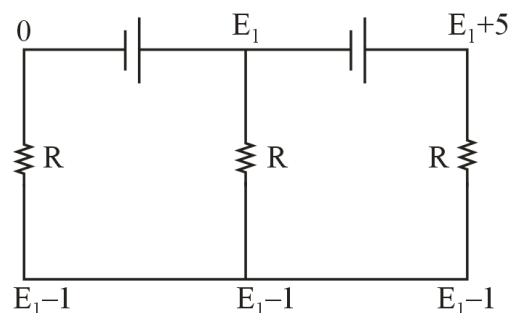
5. **Ans (3.24)**

$$\frac{1}{\lambda_o} = \frac{3}{4} R_{\text{hc}} (z_1 - 1)^2 \quad z_1 = 82$$

$$\frac{1}{n\lambda_o} = \frac{3}{4} R_{\text{hc}} (z_2 - 1)^2 \quad z_2 = 46$$

$$\therefore n = \left(\frac{z_1 - 1}{z_2 - 1} \right)^2 = \left(\frac{82 - 1}{46 - 1} \right)^2 = \left(\frac{81}{45} \right)^2 = \left(\frac{9}{5} \right)^2 = 3.24$$

6. **Ans (8.00)**



By KCL

$$\frac{E_1 - 1 - 0}{R} + \frac{E_1 - 1 - E_1}{R} + \frac{E_1 - 1 - (E_1 + 5)}{R} = 0$$

$$\Rightarrow E_1 - 1 - 1 - 1 - 5 = 0$$

$$E_1 = 8 \text{ Volts}$$

7. **Ans (12.00)**

$$Mg = \frac{\pi(d_2^2 - d_1^2)}{4} \rho \times g\ell$$

$$F = \frac{3Mg}{4} + \pi d_1 \sigma + \pi d_2 \sigma$$

$$= \pi \left[\frac{3}{16} (d_2^2 - d_1^2) \rho g\ell + (d_1 + d_2) \sigma \right]$$

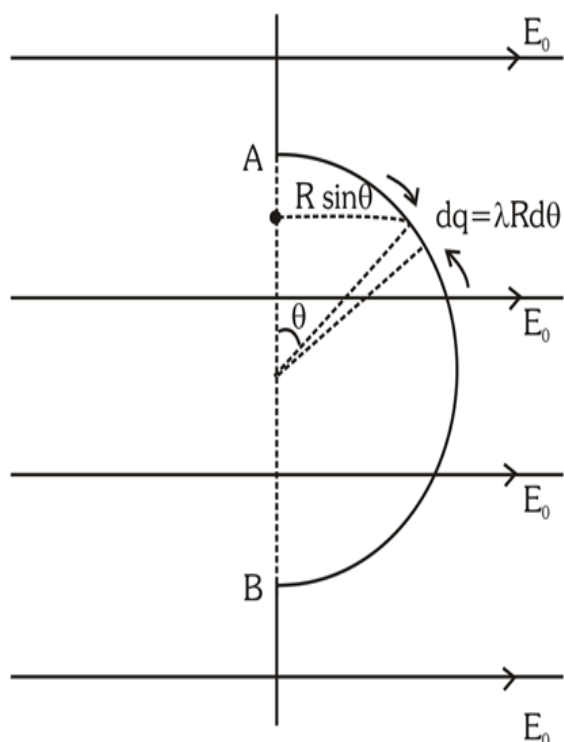
$$= \pi \left[\frac{3}{16} [16 - 1] \right] \times 10^{-6} \times 2 \times 10^3 \times 10 \times 6 \times 10^{-2} + [5 \times 10^{-3} \times 7.5 \times 10^{-2}]$$

$$= \pi \times 10^{-5} \left[\frac{45 \times 120}{16} + 37.5 \right]$$

$$= \pi \times 10^{-5} \times 375 = 3.2 \times 375 \times 10^{-5}$$

$$= 12 \text{ mV}$$

8. Ans (32.00)



$dW_{\text{Ext}} = W.$ done by Ext. agent on segment subtending

angle ' $d\theta$ '

$$= q\Delta V$$

$$= (\lambda R d\theta) (E_0 R \sin\theta)$$

$$= \lambda E_0 R^2 \sin\theta d\theta$$

$$= \lambda E_0 R^2 [-\cos\theta]_0^\pi$$

$$= 2\lambda E_0 R^2 = 2 \left(\frac{Q}{\pi R} \right) E_0 R^2 = \frac{2QE_0 R}{\pi}$$

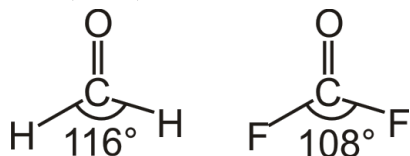
$$= \frac{2 \times (4\pi\mu C) \times (2N/C) (2m)}{\pi}$$

$$= 32\mu J$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (C,D)

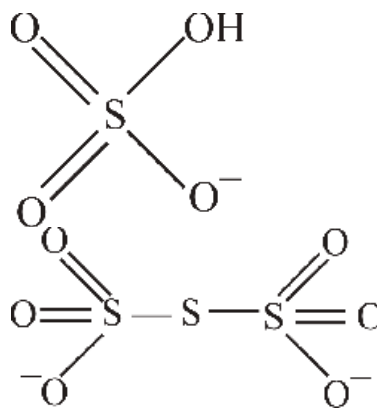


PART-2 : CHEMISTRY

SECTION-II

3. Ans (9.00)

Count the total X-O bonds



Ans. 3 & 6

Due to Resonance

4. Ans (11.00)

$$TC = T_A \left[\frac{P_A}{P_C} \right]^{\frac{1-\gamma}{\gamma}} = 400 \left[\frac{2}{1} \right]^{\frac{1-1.5}{1.5}} = \frac{400}{2^{1/3}}$$

$$\eta = - \frac{W_{\text{Total}}}{Q_{\text{absorbed}}}$$

$$= \frac{- \left[-RT_A \ln \frac{2V_0}{V_0} - R(T_C - T_B) + \frac{R}{\gamma-1} (T_A - T_C) \right]}{RT_A \ln 2}$$

$$= \frac{- \left[-400R \ln 2 - 400R \left[\frac{1}{2^{1/3}} - 1 \right] + 2R \times 400 \left[1 - \frac{1}{2^{1/3}} \right] \right]}{400R \ln 2}$$

$$\eta = \frac{400R \ln 2 - 3R \times 400 \left[1 - \frac{1}{2^{1/3}} \right]}{400R \ln 2}$$

$$\eta = 1 - \left[\frac{3 \left(1 - \frac{1}{2^{1/3}} \right)}{\ln 2} \right] = 11\%$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,D)**

$$I = \frac{1}{2} \int_0^\infty \frac{\sin(a+b)x + \sin(a-b)x}{x} dx$$

Now

$$\int_0^\infty \frac{\sin(a+b)x dx}{x} = \int_0^\infty \frac{\sin t \cdot \frac{1}{a+b} dt}{\frac{1}{a+b} \cdot t} = A$$

and

$$\int_0^\infty \frac{\sin(a-b)x dx}{x} = \int_0^\infty \frac{\sin t \cdot \frac{1}{a-b} dt}{\frac{1}{a-b} \cdot t}$$

$$= A \text{ if } a > b > 0$$

Similarly

$$\int_0^\infty \frac{\sin(a-b)x dx}{x} = - \int_0^\infty \frac{\sin(b-a)x dx}{x}$$

$$= -A \text{ if } b > a > 0$$

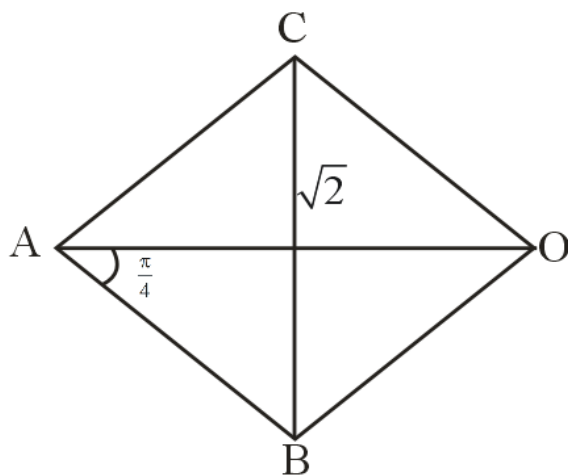
$$\Rightarrow I = \frac{A}{2} + \frac{A}{2} \text{ if } a > b > 0$$

$$\text{and } I = \frac{A}{2} - \frac{A}{2} \text{ if } b > a > 0$$

2. **Ans (A,B,C,D)**

$$\therefore OA + OB = 4, \Rightarrow OA \cdot OB \leq 4$$

$$\Rightarrow \text{area}(\triangle AOB) = \frac{1}{2} OA \cdot OB \cdot \frac{1}{\sqrt{2}} \leq \sqrt{2}$$



$$V = \frac{1}{3} \times \text{area}(\triangle AOB) \times h \leq \frac{2}{3}$$

Where h is altitude from C to base AOB. As equality is holding so $OA = OB = 2$, $h = \sqrt{2}$

so BC is altitude on base OAB from C.

$$\cos 45^\circ = \frac{2^2 + 2^2 - AB^2}{2 \cdot 2 \cdot 2}$$

3. **Ans (B,C,D)**

$$f'(x) = \cos \frac{1}{x} + \frac{1}{x} \sin \frac{1}{x}$$

$$\Rightarrow \lim_{x \rightarrow \infty} f'(x) = 1$$

$$f''(x) = \frac{-1}{x^3} \cos \frac{1}{x} < 0 \forall x \in [1, \infty)$$

$$\Rightarrow f'(x) > \lim_{x \rightarrow \infty} f'(x) = 1$$

$$f'(x) = \frac{f(x+2) - f(x)}{x+2-x}$$

$$\Rightarrow f(x+2) - f(x) > 2$$

4. **Ans (A,B,D)**

Equation of circles are

$$x^2 + y^2 - 2x - 4y + 5 - r^2 = 0 \quad \dots(1)$$

$$\text{and } x^2 + y^2 - 2x - 4y + 5 - 4r^2 = 0 \quad \dots(2)$$

equation of common chord of given circle and circle (2)

$$\text{is } 4x - 3y - 7 + 4r^2 = 0$$

$$\Rightarrow \frac{|4 - 6 - 7 + 4r^2|}{5} = r \Rightarrow 4r^2 - 9 = \pm 5r$$

$$\Rightarrow r = \frac{9}{4}, 1.$$

5. **Ans (B,C,D)**

$$y'(x) = 3y + 18 + 14x \quad y' + \left(\frac{-3}{x}\right)y = \left(\frac{18+14x}{x}\right)$$

$$\text{IF} = e^{\int \frac{-3}{x} dx} = \frac{1}{x^3}$$

$$\frac{y}{x^3} = \int \left(\frac{18}{x^4} + \frac{14}{x^3}\right) dx + C$$

$$\frac{y}{x^3} = -\frac{6}{x^3} - \frac{7}{x^2} + C$$

$$y = Cx^3 - 7x - 6 = f(1)$$

$$f(1) = C - 7 - 6 = -12 \Rightarrow C = 1$$

$$f(x) = x^3 - 7x - 6.$$

6. Ans (A, B, C)

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{bx + acx^2} + bx}{x}$$

$$\lim_{x \rightarrow -\infty} \frac{bx + acx^2 - bx^2}{x(\sqrt{bx + acx^2} - bx)}$$

$$= \lim_{x \rightarrow -\infty} \frac{\frac{b}{x} + ac - b^2}{-\sqrt{\frac{b}{x} + ac} - b}$$

$$= \frac{ac - b^2}{-(\sqrt{ac} + b)} = b - \sqrt{ac}$$

(A) $\because b$ is A.M. of a and c

$$\Rightarrow b > \sqrt{ac} \text{ (AM > GM)}$$

(B) $\because b$ is H.M. of a and c

$$\Rightarrow b < \sqrt{ac} \text{ (HM < GM)}$$

(C) $\because b = \sqrt{ac} \Rightarrow L = 0$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

p^2, q^2, r^2 are roots of equation

$$t^3 - 4t^2 - 4t - 25 = 0$$

$$\Rightarrow t^3 - 4t^2 - 4t - 25 = (t - p^2)(t - q^2)(t - r^2),$$

take log & differentiate w.r.t. t

$$\Rightarrow \frac{3t^2 - 8t - 4}{t^3 - 4t^2 - 4t - 25} = \frac{1}{t - p^2} + \frac{1}{t - q^2} + \frac{1}{t - r^2}$$

$$\Rightarrow \sum \frac{1}{4 - p^2} = \frac{48 - 32 - 4}{64 - 64 - 16 - 25} = \frac{-12}{41};$$

$$\because p^2 + q^2 + r^2 = 4 \quad \therefore 4 - p^2 = q^2 + r^2$$

$$\Rightarrow \sum \frac{p^2}{q^2 + r^2} = -3 - \frac{48}{41} = \frac{-171}{41}$$

8. Ans (A)

Put $x^3 = y$ in given equation

$$\therefore x = y^{1/3}$$

Now simplify and take cube

Therefore p^3, q^3, r^3 are roots of equation

$$y^3 - 7y^2 - 69y - 125 = 0$$

9. Ans (B)

$$\frac{z_1 - 2}{z_1 + 2} = \overline{\left(\frac{z_1 - 2}{z_1 + 2}\right)}$$

$\Rightarrow z_1$ is purely real

$$\therefore z_1 = 2$$

$$|z_1| = 2 \text{ and } \frac{z_1}{z_2} = \frac{2 \pm \sqrt{-20}}{6} = \frac{1 \pm \sqrt{5}i}{3}$$

$$\Rightarrow |z_2| = \sqrt{6}$$

$$\Rightarrow e = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{1 - \frac{4}{6}} = \frac{1}{\sqrt{3}}$$

10. Ans (A)

$$\text{Area of } \triangle OPQ = \frac{1}{2} |z_1| |z_2| \sin \theta$$

$$= \frac{1}{2} 2\sqrt{6} \frac{\sqrt{5}}{\sqrt{6}} = \sqrt{5}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (2.71 or 2.72)

$$g(x) = e^{\lim_{r \rightarrow 0} \frac{(x+1)^{r+1} - x^{r+1} - 1}{r}}$$

$$e^{\lim_{r \rightarrow 0} \frac{(x+1)^{r+1} \log(x+1) - x^{r+1} \log x}{1}} \quad (\text{by applying LH Rule})$$

$$= e^{(x+1) \ln(x+1) - x \ln x}$$

$$= \frac{(x+1)^{x+1}}{x^x}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{g(x)}{x} = \lim_{x \rightarrow \infty} \frac{x+1}{x} \cdot \left(\frac{x+1}{x}\right)^x$$

$$= 1.e = e$$

2. Ans (8.00)

$$\cos^2 \frac{(k-1)\pi}{2n} - \cos^2 \frac{k\pi}{2n} = \sin \frac{(2k-1)\pi}{2n} \sin \frac{\pi}{2n}$$

$$\Rightarrow \sin \frac{(2k-1)\pi}{2n} = \frac{1}{\sin \frac{\pi}{2n}} \left(\cos^2 \frac{(k-1)\pi}{2n} - \cos^2 \frac{k\pi}{2n} \right)$$

$$a_n = \frac{1}{\sin \frac{\pi}{2n}} \cdot \sum_{k=1}^{n-1} \left[\frac{1}{\cos^2 \frac{k\pi}{2n}} - \frac{1}{\cos^2 \frac{(k-1)\pi}{2n}} \right]$$

$$\Rightarrow a_n = \frac{1}{\sin \frac{\pi}{2n}} \left(\frac{1}{\cos^2 \frac{(n-1)\pi}{2n}} - 1 \right)$$

$$\Rightarrow a_n = \frac{1}{\sin^3 \frac{\pi}{2n}} - \frac{1}{\sin \frac{\pi}{2n}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{a_n}{n^3} = \frac{8}{\pi^3}$$

3. Ans (2.66 or 2.67)

$$\text{Since } x^4 + y^4 + z^4 + 1 = 4xyz$$

But by A.M./G.M. Inequality

$$\frac{x^4 + y^4 + z^4 + 1}{4} \geq xyz$$

$$\text{so } x^4 = y^4 = z^4 = 1 \text{ and } xyz > 0$$

so vertices are (1,1,1), (1,-1,-1), (-1,1,-1),

(-1,-1,1)

$$\text{Volume} = \frac{1}{6} \left[\begin{vmatrix} \vec{AB} & \vec{AC} & \vec{AD} \end{vmatrix} \right]$$

$$\text{Volume} = \frac{1}{6} \left| \begin{vmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{vmatrix} \right| = \frac{8}{3}$$

4. Ans (1.00)

$$A = \int_0^{\pi/2} \frac{(1 - \cos^n \theta) \sin \theta d\theta}{1 - \cos \theta} = - \int_1^0 \frac{(1 - t^n) dt}{1 - t}$$

$$= \int_0^1 (1 + t + t^2 + \dots + t^{n-1}) dt = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$$

$$B = \int_0^1 \sum_{r=1}^n n C_r (-x)^{r-1} dx$$

$$= \int_0^1 \frac{-1}{x} [(1-x)^n - 1] dx = \int_0^1 \frac{(1-x)^n - 1}{(1-x) - 1} dx$$

$$= \int_0^1 \frac{t^n - 1}{t - 1} dt = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$$

5. Ans (0.00)

$$\text{Let } P_K = \text{trace}(A^K) \text{ and Let } A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

characteristics equation : $A^3 - P_1 A^2 + KA - |A|I = O$

$$\text{where } K = \text{coeff. of } \lambda \text{ in } \begin{bmatrix} a_1 - \lambda & b_1 & c_1 \\ a_2 & b_2 - \lambda & c_2 \\ a_3 & b_3 & c_3 - \lambda \end{bmatrix} = 0$$

$$\text{i.e. } k = a_1 b_2 + a_1 c_3 + b_2 c_3 - b_3 c_2 - a_2 b_1 - a_3 c_1$$

But trace (A) = $a_1 + b_2 + c_3$ and

$$\text{trace}(A_2) = a_1^2 + b_2^2 + c_3^2 + 2a_2 b_1 + 2a_3 c_1 + 2b_3 c_2$$

$$\Rightarrow k = \frac{(\text{trace}(A))^2 - \text{trace}(A^2)}{2} = \frac{p_1^2 - p_2}{2} = \frac{16 - 10}{2} = 3$$

$$\Rightarrow A^3 - 4A^2 + 3A - |A|I = O$$

Taking Trace both sides

$$\Rightarrow 28 - 4(10) + 3(4) = 3|A|$$

$$\Rightarrow |A| = 0$$

6. Ans (0.00)

Let (x_1, y_1) is the point of intersection

$$\Rightarrow x_1 y_1 = -c^2 \text{ and } x_1^2 = -4by_1$$

$$\text{Also } \tan \theta_1 = \frac{-y_1}{x_1} \text{ and } \tan \theta_2 = \frac{-x_1}{2b}$$

$$x_1 = (4bc^2)^{1/3}$$

$$y_1 = -\frac{1}{4b} \cdot (4bc^2)^{2/3}$$

$$\therefore 2 \tan \theta_1 + \tan \theta_2 = 2 \left(-\frac{y_1}{x_1} \right) + \left(-\frac{x_1}{2b} \right)$$

$$= \frac{2c^2}{x_1^2} - \frac{x_1}{2b}$$

$$= \frac{4bc^2 - x_1^3}{2bx_1^2}$$

$$= 0$$

7. **Ans (13.00)**

Palindrome is $x y y x$ where $x \in \{1, 2, \dots, 8, 9\}$

and $y \in \{0, 9\}$

Now, $x y y x$

$$= 1000x + 100y + 10y + x = 1001x + 110y$$

Now, 1001 is divisible by 7, hence x can be anything from 1 to 9.

For 110y to be divisible by 7, y must be either 0 or 7.

$$\text{Hence, } n(S) = 10 ; n(A) = 2$$

$$\text{Probability} = \frac{2}{10} \equiv \frac{p}{10}$$

$$\Rightarrow p = 2$$

$$\text{Probability that it is divisible by 11 is } 1 \equiv \frac{q}{10}$$

$$\Rightarrow q = 10$$

$$\text{Probability that it is divisible by 13 is } \frac{1}{10} \equiv \frac{r}{10}$$

$$\Rightarrow r = 1 \text{ (} y = 0, y \text{ cannot be 13)}$$

$$\text{Hence } p + q + r = 2 + 10 + 1 = 13. \text{ **Ans.**}$$

8. **Ans (5.00)**

Let E_1 : 1st dice selected twice

$$E_1 = \{(3, 7) (7, 3) (4, 6) (6, 4) (5, 5)\} \quad P(E_1) = \frac{5}{36}$$

Let E_2 : 2nd dice selected twice

$$E_2 = \{(4, 6) (6, 4) (5, 5)\} \quad P(E_2) = \frac{3}{36}$$

E_3 : 1st drawn 1st dice and 2nd drawn 2nd dice

$$E_3 = \{(2, 8), (3, 7), (4, 6), (5, 5), (6, 4)\} \quad P(E_3) = \frac{5}{36}$$

E_4 : 1st draw 2nd dice and 2nd draw 1st dice

$$E_4 = \{(8, 2) (7, 3) (6, 4) (5, 5) (4, 6)\} \quad P(E_4) = \frac{5}{36}$$

$P(\text{same dice rolled twice / sum is 10})$

$$= \frac{\frac{1}{2} \times \frac{1}{2} \left(\frac{5}{36} + \frac{3}{36} \right)}{\frac{1}{2} \times \frac{1}{2} \left(\frac{5}{36} + \frac{3}{36} + \frac{5}{36} + \frac{5}{36} \right)} = \frac{8}{18} = \frac{4}{9} \text{ **Ans.**}$$

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE-II)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A	A,B,C,D	A,B,C	A,B,C	B,D	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	5.55 to 5.56	78.00	8.00	1.25	10.00	300.00	1.52 to 1.53

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	B,C	A,B,D	A,B,C	B,C,D	A,B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	B	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	16.00	0.50	25.00	85.00	250.00	96.00	0.84	14.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C	B,C	C	A,B,C	B,C	B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	A	C	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4000.00	4.00	375.00	2187.00	18.00	7.00	40.00	20.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A)

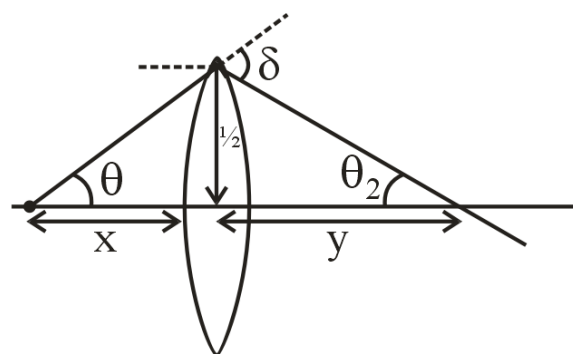
It is advisable to solve this problem in a reference frame fixed to the spiral. Using work energy theorem

$$m(v'^2 - v_0^2) = \int m\omega^2 (\vec{r} \cdot d\vec{r}) = \int m\omega^2 \rho \cdot d\rho$$

$$\frac{m(v'^2 - v_0^2)}{2} = \frac{m\omega^2 \rho^2}{2}$$

$$\Rightarrow v' = \sqrt{v_0^2 + \rho^2 \omega^2}$$

2. Ans (A,B,C,D)



$$\frac{1}{y} + \frac{1}{x} = \frac{1}{10}$$

$$\delta = \theta_1 + \theta_2 = \frac{1}{2x} + \frac{1}{2y} = \frac{1}{20} \text{ rad.}$$

$$\theta_2 = \frac{1}{2y} = \frac{1}{20} \text{ rad.}$$

3. Ans (A,B,C)

$$V_1 + V_2 = 2V$$

$$V' = 2V - V_2$$

$$V'' = 2V - V_1$$

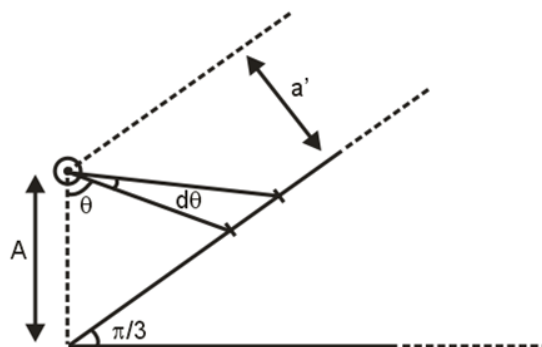
$$V' + V'' = 2V$$

$$V < 2V - 2V - V'' < 2V$$

$$V' = 2V - 2V < V'' - 2V < -V$$

$$0 < V'' < V$$

5. Ans (B,D)



$$\int \vec{B} \cdot d\vec{\ell} = \int_{AB} \vec{B} \cdot d\vec{\ell} + \int_{BC} \vec{B} \cdot d\vec{\ell}$$

$$\int \frac{\mu_0 i}{2\pi a' \sec \alpha} a' \sec \alpha dx + \int_0^{\pi/2} \frac{\mu_0 i}{2\pi a \sec \theta} a \sec \theta d\theta$$

$$= \int_0^{\pi/2} \frac{\mu_0 i}{2\pi a' \sec \alpha} a' \sec \alpha dx + \int_0^{\pi/2} \frac{\mu_0 i}{2\pi a \sec \theta} a \sec \theta d\theta$$

$$= + \frac{\mu_0 i}{6}$$

6. Ans (A,B,C)

Potential difference is defined only for electrostatic field.

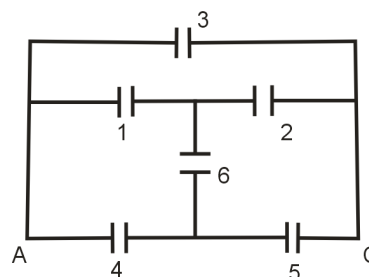
PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

For terminal A & C circuit can be redrawn as wheat stone bridge is balanced.

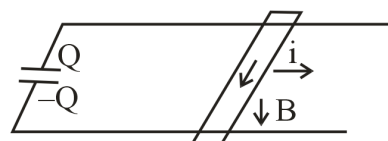
Hence charge on C_6 is zero.



PART-1 : PHYSICS

SECTION-II

1. Ans (5.00)



$$i\ell B = m \frac{dv}{dt}$$

$$mv = B\ell(Q_0 - Q)$$

In steady state

$$Q = C B \ell u$$

$$\Rightarrow u = \frac{B\ell Q_0}{m + B^2 \ell^2 C} = \frac{4 \times 2.5 \times 10^{-3}}{2 \times 10^{-3}} = 5 \text{ m/s}$$

2. Ans (5.55 to 5.56)

$$R_\infty = \frac{m_e z^2 e^4}{8 \epsilon_0^2 n^2 h^2}$$

$$R_H = \frac{m m_e}{m + m_e} \frac{z^2 e^4}{8 \epsilon_0^2 n^2 h^2}$$

$$\frac{R_\infty - R_H}{R_\infty} = \frac{m_e - \frac{m m_e}{m + m_e}}{m_e} = 1 - \frac{m}{m + m_e}$$

$$= \frac{m_e}{m + m_e} = \frac{\frac{1}{1800}}{1 + \frac{1}{1800}}$$

$$= \frac{1}{1800} \times 10^4 = 5.55$$

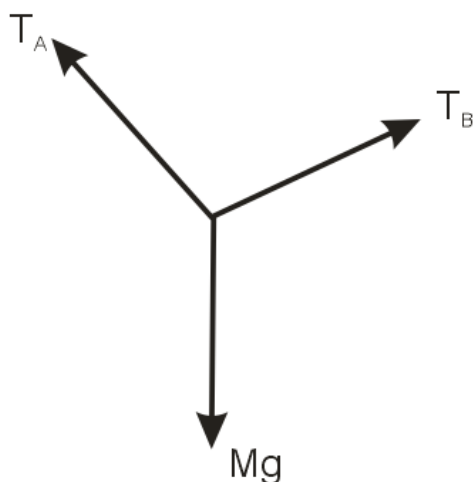
3. **Ans (78.00)**

$$\vec{T}_A + \vec{T}_B + \vec{mg} = 0 \Rightarrow m = 5\text{kg},$$

$$\frac{T_A'}{T_B'} = \frac{T_A}{T_B} = \frac{4}{3}$$

$$\vec{T}_A + \vec{T}_B + \vec{mg} + F = 0 \Rightarrow T_A'^2 + T_B'^2 = 150^2$$

$$\Rightarrow T_B = 78\text{N}$$



4. **Ans (8.00)**

$$\text{At } r_y \rightarrow \infty, V_R : 16V$$

5. **Ans (1.25)**

$$\frac{d_I}{i} = \frac{d_0}{\mu} = \frac{h_3 - h_1}{\mu}$$

$$d_I = h_3 - h_2$$

$$\Rightarrow \mu = \frac{h_3 - h_1}{h_3 - h_2} = \frac{4.80}{2.40} = 2$$

$$\frac{du}{\mu} = \frac{dh_3 + dh_1}{h_3 - h_1} + \frac{dh_3 + dh_2}{h_3 - h_2}$$

$$= \frac{0.02}{4.80} + \frac{0.02}{2.40}$$

$$= \frac{0.02[3]}{4.80} \times 100$$

$$\% \text{ error} = \frac{60}{48}$$

6. **Ans (10.00)**

There is no horizontal force, momentum is conserved

$$mv_1 - mv_2 = 0$$

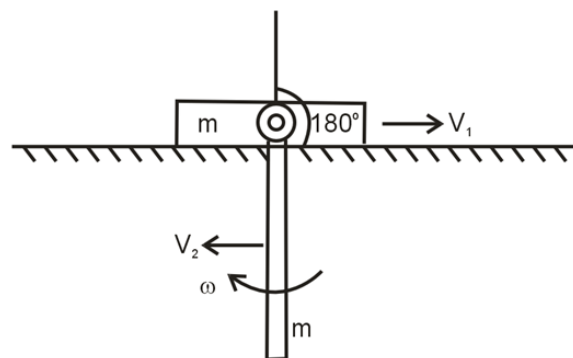
for hinged point

$$v_1 = \frac{L}{2} \omega - v_2$$

Energy conservation,

$$mgL = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 + \frac{1}{2} \frac{mL^2 \omega^2}{12}$$

$$\text{Solving, } v_1 = \sqrt{\frac{3}{5}}gL, v_1 = 10 \text{ m/s.}$$



7. **Ans (300.00)**

Power absorbed by earth = power emitted by earth

$$\Rightarrow \frac{e\sigma (4\pi R_s^2) T_s^4}{4\pi r^2} \times \pi R_e^4 = e\sigma (4\pi R_e^2) T_e^4$$

$$\Rightarrow T_e = T_s \sqrt{\frac{R_s}{2r}} = T_s \sqrt{\frac{R_s}{2 \times 200 R_s}} \Rightarrow T_e = \frac{T_s}{20} = 300\text{K}$$

8. **Ans (1.52 to 1.53)**

The center of mass of the rod is at distance

$$\frac{(y-x)}{2} \text{ from brim.}$$

$$\left(\rho_{al} gV - \rho_w g \frac{V}{2} \right) x = mg \frac{(y-x)}{2}$$

$$\Rightarrow \frac{4}{x} = 1.53$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A,B,C)**

Electron with drawing group on alkene strengthen back bonding

π -acidic CO weakens metal-alkene back bonding

3. **Ans (A,B,D)**

(A) Adiabatic since heat energy will flow from high to low temperature. spontaneously therefore it is adiabatic it is adiabatic irreversible process. $\Delta S_{\text{sys}} > 0$

(B) same reason as of (A) option

(C) If $T_A = T_B$

$$\Delta S = \Delta S_A + \Delta S_B = 0 + 0 = 0$$

(D) $W = 0$ $Q = 0$ $\Delta U = 0$

4. **Ans (A,B,C)**

C-2 epimers give same osazone

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (16.00)**

%SN² reaction

$$= \frac{4.7 \times 10^{-5} [\text{RX}][\text{OH}] \times 100}{4.7 \times 10^{-5} [\text{RX}][\text{OX}] + 2.4 \times 10^{-6} [\text{RX}]} = \frac{4.7[\text{OH}] \times 100}{4.7[\text{OH}] + 0.24}$$

$$= \frac{4.7 \times 0.01 \times 100}{4.7 \times 0.01 + 0.24} = \frac{4.7}{0.047 + 0.24} = \frac{4.7}{.287} = 16.38$$

2. **Ans (0.50)**

$[\text{Ag}^+]$, when AgCl starts precipitating

$$= \frac{1.6 \times 10^{-10}}{0.2} = 8 \times 10^{-9}$$

$$[\text{At that time } [\text{Br}^-] = \frac{8 \times 10^{-13}}{8 \times 10^{-9}} = 10^{-4}$$

The percent of Br⁻ remaining in the solution

$$= \frac{10^{-4}}{0.2} \times 100 = 0.5\%$$

3. **Ans (25.00)**

Anions occupy fcc positions and half of the tetrahedral holes and 8 tetrahedral holes per unit cell, the fraction of volume occupied by spheres per unit volume of the unit cell is

$$= \frac{4 \times \left(\frac{4}{3}\pi r_a^3\right) + \frac{1}{2} \times 8 \times \left(\frac{4}{3}\pi r_c^3\right)}{16\sqrt{2}r_a^3} = \frac{\pi}{3\sqrt{2}} \left\{ 1 + \left(\frac{r_c}{r_a}\right)^3 \right\}$$

∴ for tetrahedral holes,

$$\frac{r_c}{r_a} = 0.225$$

$$= \frac{\pi}{3\sqrt{2}} \{ 1 + (0.225)^3 \} = 0.7493$$

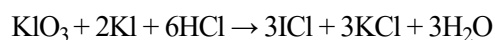
∴ Void volume = 1 - 0.7493

$$= 0.2507 / \text{unit volume of unit cell.}$$

$$\% \text{ void space} = 25.07\%$$

4. **Ans (85.00)**

It is given that 20 mL of stock solution of KI requires 30 mL of M/10 KIO₃ solution to convert I⁻ ions into ICl according to the reaction



For 50 mL of KI solution, the required volume of KIO₃ solution would be

$$\frac{30\text{mL of KIO}_3 \text{ solution}}{20\text{mL of KI solution}} \times 50\text{mL of KI}$$

Solution = 75 mL of KIO₃ solution

After treating 50 mL of KI solution with 1 g AgNO₃

Sample, the volume of KIO₃ solution used is 50 mL.

This means KI equivalent to 25 mL of KIO₃ solution is used in precipitating out Ag⁺ ions from 50 mL of KI solution.

Amount of KIO₃ in 25 mL of M/10KIO₃ solution (25 mL)

$$\left(\frac{1}{10} \text{M}\right) = \left(\frac{25}{100} \text{L}\right) \left(\frac{1}{10} \text{mol L}^{-1}\right) = 0.0025 \text{ mol}$$

Amount of KI equivalent to this amount

$$= 2 \times 0.0025 \text{ mol}$$

Mass of AgNO₃ precipitated out

$$= 2 \times 0.0025 \times 170 \text{ g} = 0.85 \text{ g}$$

$$\text{Percent of AgNO}_3 \text{ in the sample} = \frac{0.85\text{g}}{1.0\text{g}} \times 100 = 85$$

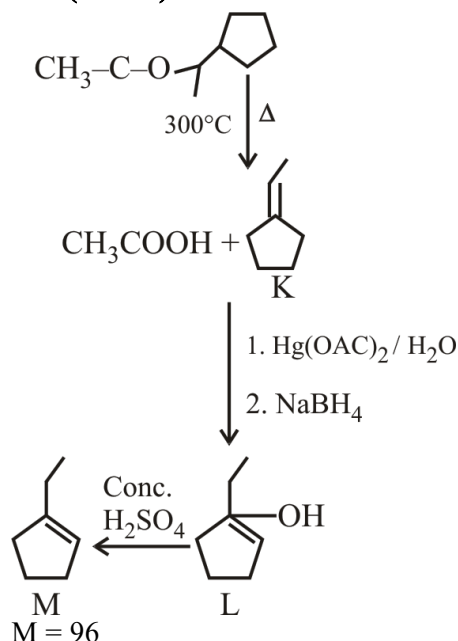
5. **Ans (250.00)**

0.25 gm prevents coagulation of 10 ml of gold sol by 1 ml of 10% NaCl

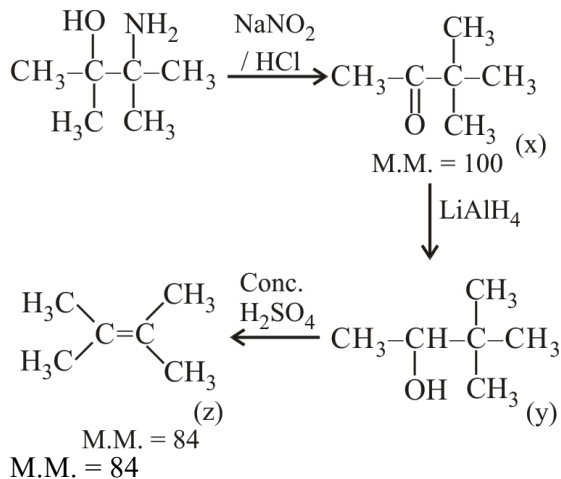
∴ milligrams at starch = 250

∴ Gold number of = 250

6. **Ans (96.00)**



7. Ans (0.84)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,C)

$$\text{We have } f(x) = x + x \int_{-1}^1 f(t) dt - x^2 \int_{-1}^1 t f(t) dt$$

$$f(x) = (1+A)x - Bx^2 \dots (1)$$

$$\text{where } A = \int_{-1}^1 f(t) dt \text{ and } B = \int_{-1}^1 t f(t) dt$$

$$\text{Now } A = \int_{-1}^1 f(t) dt = \int_{-1}^1 ((1+A)t - Bt^2) dt$$

$$\Rightarrow A = -2B \int_0^1 t^2 dt$$

$$\therefore A = \frac{-2B}{3} \dots (2)$$

$$\text{Now } B = \int_{-1}^1 t f(t) dt = \int_{-1}^1 t ((1+A)t - Bt^2) dt$$

$$\Rightarrow B = 2(A+1) \int_0^1 t^2 dt;$$

$$\therefore B = \frac{2}{3}(A+1) \dots (3)$$

$$\text{On solving (2) and (3), we get } A = \frac{-4}{13}, B = \frac{6}{13}$$

$$\therefore \text{ from equation (1), we get, } f(x) = \frac{9}{13}x - \frac{6}{13}x^2$$

$$\Rightarrow f'(x) = \frac{9}{13} - \frac{12}{13}x$$

$$\therefore f'\left(-\frac{1}{3}\right) = \frac{9}{13} - \frac{12}{13}\left(-\frac{1}{3}\right) = \frac{9}{13} + \frac{4}{13} = \frac{13}{13} = 1.$$

2. Ans (B,C)

(A) If for some value of x say $x = x_1$, the equation $E = 0$ has one root infinite and other root a non zero finite, so $u = 0$ and $v \neq 0$

$\therefore u = 0$ and $v = 0$ do not have a common root.

$\Rightarrow$ (A) is False

(B) If for some value of x say $x = x_2$, the equation $E = 0$ has both root infinite root, so $u = 0$ and $v = 0$

$\therefore u = 0$ and $v = 0$ will have a common root.

$\Rightarrow$ (B) is True

(C) If for some value of x say $x = x_3$, the equation $E = 0$ becomes an identity in y , so $u = 0$, $v = 0$ and $w = 0$

$\therefore u = 0$, $v = 0$ and $w = 0$ must have a root, common to all of them.

$\Rightarrow$ (C) is true

(D) If for some value of x say $x = x_4$, the equation $E = 0$ has both roots real and distinct then $v = 0$ and $w = 0$ must have a common root is not necessary

$\Rightarrow$ (D) is False

3. Ans (C)

$$2ax + 2byy' = 0$$

$$y' = \frac{-ax}{by}$$

$$y'' = \frac{-aby + abxy'}{b^2y^2}$$

$$y^2 \frac{y''}{ab} = \frac{xy' - y}{b^2}$$

$$\frac{b}{a}y^2y'' = x \left(\frac{-ax}{by} \right) - y$$

$$\frac{b}{a}y^2y'' + \frac{ax^2}{by} + y = 0$$

$$\frac{b}{a}y^3y'' + \frac{ax^2}{b} + y^2 = 0$$

$$\frac{b}{a}y^3y'' + \frac{c}{b} = 0$$

$$y^3y'' + \frac{ac}{b^2} = 0$$

$$\therefore \frac{ac}{b^2} = 4 \Rightarrow 4b^2 = ac$$

$$ax^2 + 4bx + c = 0$$

$$D = 16b^2 - 4ac$$

$$= 4(4b^2 - ac) = 0$$

real and equal roots

4. Ans (A,B,C)

$$f(x) = \sin x + ax + b$$

if $a > 1$ then $f(x)$ is increasing

so only one real root, which is positive if $b > 0$

and negative if $b < 0$

if $a < -1$

$f(x)$ is decreasing so only one real root, which

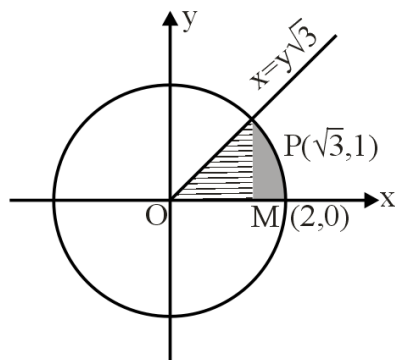
is negative if $b < 0$

5. Ans (B,C)

$$f(x) = \cos(\sin x)$$

range is $[\cos 1, 1]$

6. Ans (B)



The line $y = x\sqrt{3}$ meets the circle in the first

quadrant at $P(\sqrt{3}, 1)$. Therefore

$$\begin{aligned} \text{Area required} &= \Delta OPM + \int_{\sqrt{3}}^2 \sqrt{4-x^2} dx \\ &= \frac{1}{2} \times \sqrt{3} \times 1 + \left[\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right]_{\sqrt{3}}^2 \\ &= \frac{\sqrt{3}}{2} + \left[2\sin^{-1} 1 - \frac{\sqrt{3}}{2} - 2\sin^{-1} \frac{\sqrt{3}}{2} \right] \\ &= \frac{\sqrt{3}}{2} + \pi - \frac{\sqrt{3}}{2} - 2 \times \frac{\pi}{3} \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

$$(P) 2n!n! = 1152 \Rightarrow (n!)^2 = 576 \Rightarrow n! = 24$$

$$\Rightarrow n = 4 \text{ Ans.}$$

$$(Q) \text{ Let } a^2 + b^2 = x$$

$$1 - 2ab = (a+b)^2 - 2ab = a^2 + b^2 = x$$

$$(a+b=1); \text{ also } ab = \frac{1-x}{2}$$

$$\text{and } a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$$

$$a^3 + b^3 = x - ab \quad \dots (1)$$

$$\text{but } ab = \frac{1-x}{2}$$

$$\text{Hence } a^3 + b^3 = x - \frac{1-x}{2} = \frac{3x-1}{2}$$

Hence the equation

$$(1-2ab)(a^3 + b^3) = 12, \text{ becomes}$$

$$x \left(\frac{3x-1}{2} \right) = 12$$

$$3x^2 - x - 24 = 0$$

$$(x-3)(3x+8) = 0$$

$$x = 3 \text{ or } x = -\frac{8}{3} \text{ (not possible) as } x = a^2 + b^2 \neq 0$$

$$\therefore x = 3 \Rightarrow a^2 + b^2 = 3 \text{ Ans.}$$

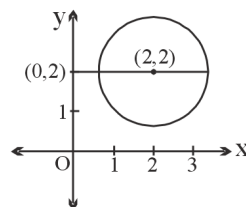
$$(R) |z - 2 - 2i| \leq 1$$

denotes the region inside a circle with centre (2, 2) and radius is 1

$$\begin{aligned} |2iz + 4| &= |2i(z-2i)| = |2i||z-2i| \\ &= 2|z-2i| \end{aligned}$$

$$|z-2i| = \text{distance of } z \text{ from } P(0, 2)$$

$$\text{Hence } M = 2 \cdot 3 = 6 \text{ Ans.}$$



$$\begin{aligned} (S) \log^2(\sin x) - \log^2|\cos x| + \frac{\log_3(\tan x) + \log_5(\tan x)}{\text{can be 0 when } x = \pi/4} &= 0 \\ \Rightarrow \log_2 \frac{\sin x}{|\cos x|} + \frac{\log_3(\tan x) + \log_5(\tan x)}{\text{can be 0 when } x = \pi/4} &= 0 \end{aligned}$$

{ For $x \in (0, \frac{\pi}{4})$, each term on L.H.S. of equation is negative and for $x \in (\frac{\pi}{4}, \frac{\pi}{2})$, each term on L.H.S. of equation is positive }

$$\text{Hence } x = \frac{\pi}{4} = \frac{-7\pi}{4} \text{ or}$$

$\therefore$ number of solutions in $(-2\pi, 2\pi)$ is 2 Ans.]

8. Ans (A)

(P) Clearly $x - x^4 + x^7 - x^8 - 1 < 0 \forall x \in \mathbb{R}^+$.

Hence $\text{sgn}(x - x^4 + x^7 - x^8 - 1) = -1 \forall x \in \mathbb{R}^+$.

(Q) We have $f(x) + x f(-x) = x + 1$ (1)

Replace x by $-x$ in (1), we get

$$-x f(x) + f(-x) = 1 - x \quad \text{..... (2)}$$

$\therefore$ From (1) and (2), we get

$$x^2 f(x) - x f(-x) = x^2 - x \quad \text{..... (3)}$$

$$\Rightarrow (1 + x^2) f(x) = 1 + x^2$$

Hence $f(x) = 1$

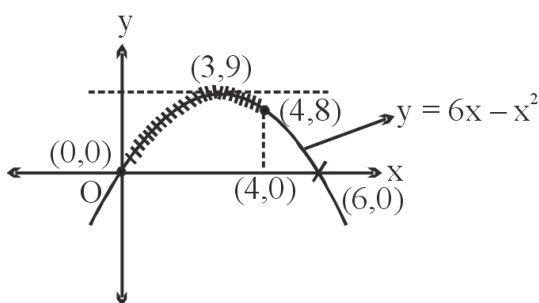
(R) We have $f(x) = \sqrt{6x - x^2}$

Clearly range of $f = [0, 3] \forall 0 \leq x \leq 4$.

(S) We have $f(x) = 2^{|x-1| + |x-2|}$

$$= \begin{cases} 2^{-2x+3} & ; 0 \leq x \leq 1 \\ 2 & ; 1 < x < 2 \\ 2^{2x-3} & ; 2 \leq x \leq 3 \end{cases}$$

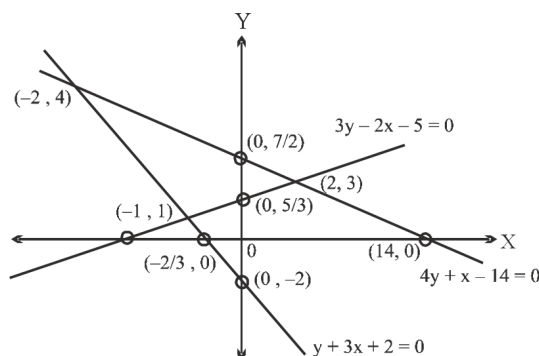
Clearly range of $f = [2, 8] \forall 0 \leq x \leq 3$.



9. Ans (C)

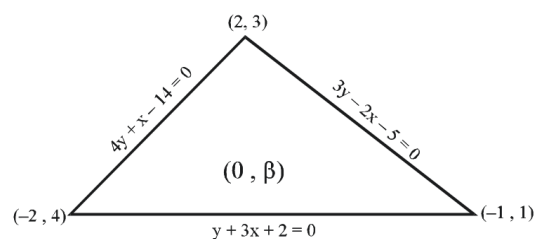
Diagrammatic solution

[Ans. $\{2, 3\}$, where $k \in \left[\frac{5}{3}, \frac{7}{2}\right]$]



Obviously $5/3 \leq k \leq 7/2$

Alternatively:



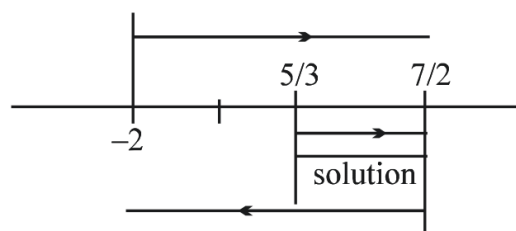
Substituting $(2, 3)$ in $y + 3x + 2 = 0$

we get $13 > 0$

$$\beta + 2 \geq 0$$

$$\text{similarly } 3\beta - 5 \geq 0 \quad 5/3 \leq \beta \leq 7/2$$

$$\text{and } (2\beta - 7) \leq 0$$



$$(Q) [\vec{V}_1 \ \vec{V}_2 \ \vec{V}_3] = \begin{vmatrix} 1 & 2 & 3 \\ 0 & \lambda & 4 \\ 0 & 0 & 2\lambda - 1 \end{vmatrix} \cdot [\hat{a} \ \hat{b} \ \hat{c}]$$

$$1(\lambda(2\lambda - 1)) [\hat{a} \ \hat{b} \ \hat{c}] \neq 0$$

$$= \lambda(2\lambda - 1) [\hat{a} \ \hat{b} \ \hat{c}] \neq 0$$

$$\text{as } [\hat{a} \ \hat{b} \ \hat{c}] \neq 0 \text{ hence } [\vec{V}_1 \ \vec{V}_2 \ \vec{V}_3] \neq 0$$

$$\Rightarrow \lambda(2\lambda - 1) \neq 0.$$

Hence $\vec{V}_1, \vec{V}_2$ and $\vec{V}_3$ are non coplanar

for all $\lambda \in \mathbb{R}$ except $\lambda = 0$ or $\lambda = \frac{1}{2} \Rightarrow Q, R, S, T$

(R) We have $\vec{n} = \hat{i} - 2\hat{j} - \hat{k}$

Equation of line through A

$$\frac{x-2}{1} = \frac{y+3}{-2} = \frac{z-3}{-1} = \lambda$$

$$x = \lambda + 2$$

$$y = -(2\lambda + 3)$$

$$z = 3 - \lambda$$

$$\text{Hence } \lambda + 2 + 2(2\lambda + 3) - (3 - \lambda) + 1 = 0$$

$$\Rightarrow 6\lambda + 6 = 0 \Rightarrow \lambda = -1$$

$$\text{Hence } Q \text{ is } (1, -1, 4)$$

$$\text{Hence } a_1 + 2 = 2 \Rightarrow a_1 = 0$$

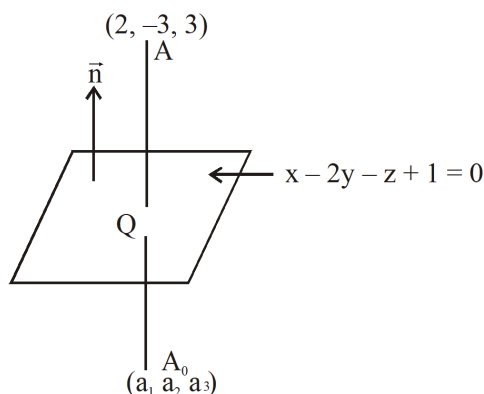
$$a_2 - 3 = -2 \Rightarrow a_2 = 1$$

$$a_3 + 3 = 8 \Rightarrow a_3 = 5$$

$$\text{Image is } (0, 1, 5)$$

Obviously distance of image of the point of z-axis

is 1 Ans. $\Rightarrow$ (S)

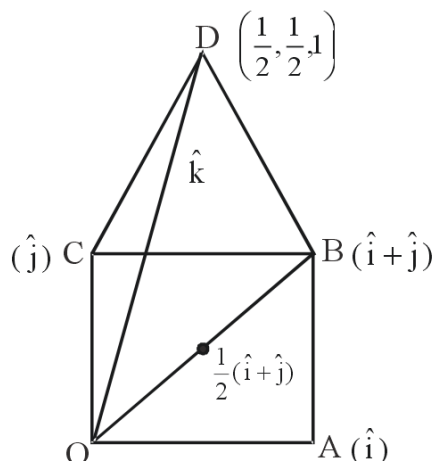


$$(S) \vec{V}_1 = \frac{1}{2}(\hat{i} + \hat{j} + 2\hat{k})$$

$$\vec{OB} = \vec{V}_2 = (\hat{i} + \hat{j})$$

$$\therefore \cos \theta = \frac{\vec{V}_1 \cdot \vec{V}_2}{|\vec{V}_1||\vec{V}_2|} = \frac{1+1}{\sqrt{2} \cdot \sqrt{6}} = \frac{1}{\sqrt{3}}$$

$$\theta = \tan^{-1} \sqrt{2} \Rightarrow K = 2 \text{ Ans.}]$$



10. Ans (D)

Here 24 matrices are possible value of

determinants can be $-8, -4, -2, 2, 4, 8$

(A) Possible non-negative values of $|A|$ are 2, 4, 8

$\Rightarrow \text{sum} = 14$

(B) Sum of these 24 determinants is 0

(C) Mod. $(\det(A))$ is least $|A| = \pm 2$

$$\Rightarrow |\text{adj}(\text{adj}(\text{adj}A))| = |A|^{(n-1)^3} = \pm 2$$

(D) Least value of $\det(A)$ is -8 Now $|4A^{-1}| = 16$

$$\frac{1}{|A|} = \frac{16}{-8} = -2$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (4000.00)

$$\text{Given } S = x^2 + 4xh = 1200$$

$$\text{and } V = x^2h$$

$$V(x) = \frac{x^2(1200 - x^2)}{4x}; V(x) = \frac{1}{4}(1200x - x^3)$$

$$\text{Put } V'(x) = 0 \text{ gives } x = 20$$

$$\text{If } x = 20, h = 10$$

$$\text{Hence } V_{\max.} = x^2h = (400)(10) = 4000 \text{ cubic cm. Ans.}$$

2. Ans (4.00)

We have

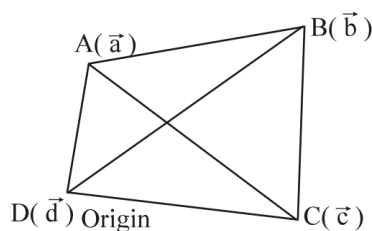
$$\vec{AB}^2 + \vec{BC}^2 + \vec{CD}^2 + \vec{DA}^2 - \vec{AC}^2 - \vec{BD}^2 = \lambda \vec{PQ}^2$$

$$(\vec{b} - \vec{a})^2 + (\vec{c} - \vec{b})^2 + (\vec{d} - \vec{c})^2 + (\vec{a} - \vec{d})^2 - (\vec{c} - \vec{a})^2 - (\vec{d} - \vec{b})^2 = \lambda (\vec{PQ})^2$$

$$\text{on simplifying gives}$$

$$= \vec{a}^2 + \vec{b}^2 + \vec{c}^2 - 2\vec{a} \cdot \vec{b} - 2\vec{b} \cdot \vec{c} + 2\vec{a} \cdot \vec{c}$$

$$(\vec{a} + \vec{c} - \vec{b})^2 \dots (1)$$



$$\text{again } \vec{P} = \frac{\vec{a} + \vec{c}}{2}; \vec{Q} = \frac{\vec{b}}{2};$$

$$\vec{PQ} = \frac{\vec{a} + \vec{c} - \vec{b}}{2} \Rightarrow 4(\vec{PQ})^2 = (\vec{a} + \vec{c} - \vec{b})^2$$

$$\Rightarrow (\vec{PQ})^2 = \frac{(\vec{a} + \vec{c} - \vec{b})^2}{4} \dots (2)$$

from (1) and (2)

$$\lambda = 4 \text{ Ans.}$$

3. Ans (375.00)

$$\text{Given } A^2 - 4A - 5I = 0$$

$$A^3 = A \cdot A^2 = A(4A + 5I) = 4A^2 + 5A$$

$$= 4(4A + 5I) + 5A = 21A + 20I$$

$$= \begin{bmatrix} 21 & 42 & 42 \\ 42 & 21 & 42 \\ 42 & 42 & 21 \end{bmatrix} + \begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{bmatrix}$$

$$= 315 + 60 = 375 \text{ Ans.}$$

4. Ans (2187.00)

$$\text{We know that } (a-1)^n = {}^nC_0 a^n - {}^nC_1 a^{n-1} + {}^nC_2 a^{n-2} - \dots + (-1)^{n-1} {}^nC_{n-1} a + (-1)^n {}^nC_n$$

$$\therefore \frac{(a-1)^n}{a} = {}^nC_0 a^{n-1} - {}^nC_1 a^{n-2} + {}^nC_2 a^{n-3} - \dots + (-1)^{n-1} {}^nC_{n-1} + (-1)^n {}^nC_n$$

$$\text{Hence } f(n) = \frac{(a-1)^n - (-1)^n}{a}$$

$$\text{Now, } f(2007) + f(2008)$$

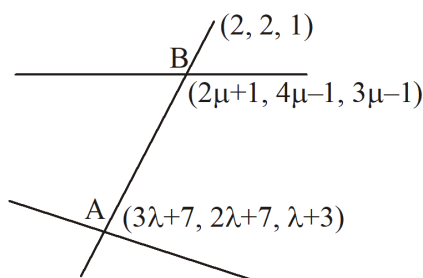
$$= \frac{(a-1)^{2007} + 1}{a} + \frac{(a-1)^{2008} - 1}{a}$$

$$= \frac{(a-1)^{2007}}{a} = \left(\frac{1}{3^{223}} \right)^{2007} = 3^9 = 3^2 \cdot 3^7 = 9(2187)$$

$$\text{Hence } k = 2187 \text{ Ans.}$$

5. Ans (18.00)

$$\frac{3\lambda - 2\mu + 6}{2} = \frac{2\lambda - 4\mu + 8}{2} = \frac{\lambda - 3\mu + 4}{1}$$



On solving we get $\mu = 0$ and $\lambda = 2$

Hence $A(13, 11, 5); B(1, -1, -1)$

$$AB = \sqrt{144 + 144 + 36} = \sqrt{324} = 18 \text{ Ans.}$$

6. Ans (7.00)

$$\text{Let } I = \int_{-1}^1 \cot^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right) \left(\cot^{-1} \frac{x}{\sqrt{1-(x^2)^{|x|}}} \right) dx \dots (1)$$

Using King

$$I = \int_{-1}^1 \left(\cot^{-1} \frac{1}{\sqrt{1-x^2}} \right) \left(\cot^{-1} \frac{-x}{\sqrt{1-(x^2)^{|x|}}} \right) dx \dots (2)$$

On adding

$$2I = \int_{-1}^1 \cot^{-1} \frac{1}{\sqrt{1-x^2}} \left\{ \cot^{-1} \frac{x}{\sqrt{1-(x^2)^{|x|}}} + \pi - \cot^{-1} \frac{x}{\sqrt{1-(x^2)^{|x|}}} \right\} dx$$

$$2I = \int_{-1}^1 \pi \cot^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right) dx = \int_{-1}^1 \pi \tan^{-1} \sqrt{1-x^2} dx$$

$$= 2\pi \int_0^1 \tan^{-1} \sqrt{1-x^2} dx \dots (3)$$

(As $\tan^{-1} \sqrt{1-x^2}$ is even function)

$$\therefore I = \pi \int_0^1 \underbrace{1}_{II} \underbrace{\tan^{-1} (\sqrt{1-x^2})}_I dx \dots (4)$$

Integrating by parts

$$I = \underbrace{\tan^{-1} (\sqrt{1-x^2})}_{\text{zero}} \cdot x \Big|_0^1 - \int_0^1 \frac{x}{(1+x^2)} \cdot \frac{(-x)}{\sqrt{1-x^2}} dx$$

$$= \pi \int_0^1 \frac{x^2}{(2-x^2)\sqrt{1-x^2}} dx$$

$$\text{Let } x = \sin \theta \Rightarrow dx = \cos \theta d\theta$$

$$I = \pi \int_0^{\pi/2} \frac{\sin^2 \theta}{(2-\sin^2 \theta)} d\theta = -\pi \int_0^{\pi/2} \frac{2-\sin^2 \theta - 2}{2-\sin^2 \theta} d\theta$$

$$\therefore I = 2\pi \int_0^{\pi/2} \frac{d\theta}{2-\sin^2 \theta} - \frac{\pi^2}{2}$$

$$= 2\pi \int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{2+2\tan^2 \theta - \tan^2 \theta} - \frac{\pi^2}{2}$$

$$= 2\pi \int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{2+\tan^2 \theta} - \frac{\pi^2}{2}$$

$$\text{put } \tan \theta = t$$

$$I = 2\pi \int_0^\infty \frac{dt}{2+t^2} - \frac{\pi^2}{2} = 2\pi \left[\frac{1}{\sqrt{2}} \tan^{-1} \frac{t}{\sqrt{2}} \right]_0^\infty - \frac{\pi^2}{2}$$

$$= \frac{\pi^2}{\sqrt{2}} - \frac{\pi^2}{2} = \frac{\pi^2 (\sqrt{2}-1)}{2} = \frac{\pi^2 (\sqrt{a}-\sqrt{b})}{\sqrt{c}}$$

$$\Rightarrow a = 2, b = 1 \text{ and } c = 4$$

$$\Rightarrow a + b + c = 2 + 1 + 4 = 7 \text{ Ans.}$$

7. **Ans (40.00)**

We have $\frac{dy}{dx} = 20(1600 - y^2)$

$$\Rightarrow \int \frac{dy}{(40)^2 - y^2} = 20 \int dx$$

$$\Rightarrow \frac{1}{80} \ln \frac{40+y}{40-y} = 20x + C' \text{ or } \ln \frac{40+y}{40-y}$$

$$= 1600x + C$$

$$\Rightarrow \frac{40+y}{40-y} = \frac{ke^{1600x}}{1} \text{ where } k = e^C \text{ (let)}$$

$$\Rightarrow \frac{2y}{80} = \frac{ke^{1600x} - 1}{ke^{1600x} + 1} \text{ (using componendo \& dividendo)}$$

$$\therefore \lim_{x \rightarrow \infty} y = 40 \lim_{x \rightarrow \infty} \left[\frac{k - e^{-1600x}}{k + e^{-1600x}} \right] = 40 \text{ Ans.}$$

8. **Ans (20.00)**

Let $t = m + w$.

$$\text{We are given that } \frac{10}{19} = \frac{mw}{tC_2} = \frac{2mw}{t(t-1)} \dots(1)$$

$$\text{But } \sqrt{mw} \leq \frac{m+w}{2} \text{ or } 2\sqrt{mw} \leq t \text{ (G.M. } \leq \text{ A.M.)}$$

$$\text{or } mw \leq \frac{t^2}{4} \dots(2)$$

$$\text{Substituting } mw = \left(\frac{10}{19} \right) \left(\frac{t(t-1)}{2} \right)$$

$$\text{We have } \left(\frac{5}{19} \right) t(t-1) \leq \frac{t^2}{4} \Rightarrow 20(t^2 - t) \leq 19t^2$$

$$\Rightarrow 20t^2 - 20t \leq 19t^2 \text{ or } t^2 \leq 20t \Rightarrow t \leq 20$$

$$\Rightarrow t_{\max} = 20$$

Also equality holds if $m = w = 10$

$\therefore$ Maximum children in class = 20 **Ans.**